

Final Exam

(Friday, Dec. 13th, 2013, 1:30-3:30pm)

Open books, open notes. Computers and cellphones should be turned off.

Student's Last Name:

Student's First Name:

Student ID:

Email:

Problem #	Points	Out of Total
1		20
2		20
3		30
3		30
Total		100

Academic Honesty Policy:

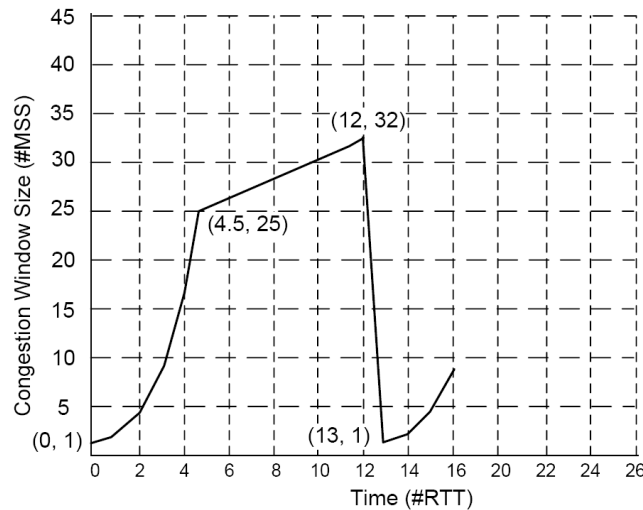
I agree to abide by the UCI Academic Senate Policy on Academic Honesty (Appendix VIII.B), which specifies that students have responsibility for:

- 1. Refraining from cheating and plagiarism.*
- 2. Refusing to aid or abet any form of academic dishonesty.*
- 3. Notifying professors and/or appropriate administrative officials about observed incidents of academic misconduct. The anonymity of a student reporting an incident of academic dishonesty will be protected.*

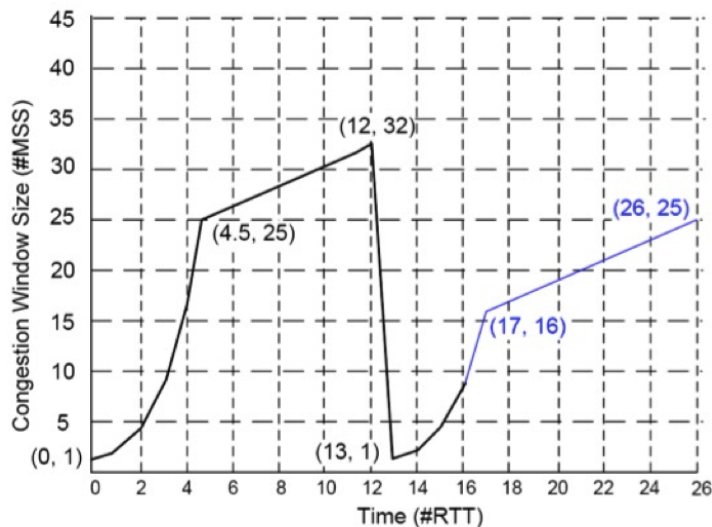
Student's Signature:

GOOD LUCK!

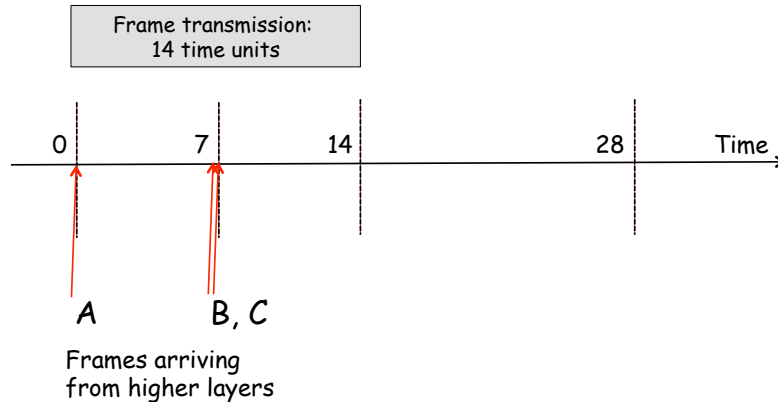
1. (20 Points) **TCP.** Consider the following plot of TCP window size as a function of time (counted in number of RTTs). The coordinate of each turning point has been marked in the plot.



- (a) (5) Identify the intervals of time when TCP slow start is operating. What is the respective SSThres (slow start threshold) in TCP congestion control? **Observe the exponential growth in the window size**
 Answer: (0, 4.5) and (13, 16). The SSThres are 25 and 16, respectively.
- (b) (2 Points) Identify the interval(s) of time when TCP congestion avoidance is operating. What is the respective SSThres?
 Answer: (4.5, 12). The SSThres is 25.
- (c) (3 Points) How many segments are lost from time 0 to 16?
 Answer: 1
- (d) (10 Points) Assume that no packet is lost during time 16-26. Complete the TCP congestion window curve from time 16 to 26.
 Answer: The critical information in the curve was the y-coordinate of the turning point in what you add, which is the indicator of the SSThres.



2. (20 Points) **Random Access Protocols.** Consider hosts A,B,C that transmit on a shared medium, using a random access protocol. Each host has exactly one frame to send, with transmission time = 14 time units. Host A wants to transmit at time $t = 0$. Hosts B,C both want to transmit at time $t = 7$, as shown in the figure below. Time is slotted and all hosts are synchronized to the clock.



If needed to make randomized decisions, each host has access to the following sequence of random numbers, drawn uniformly at random between 0 and 1:

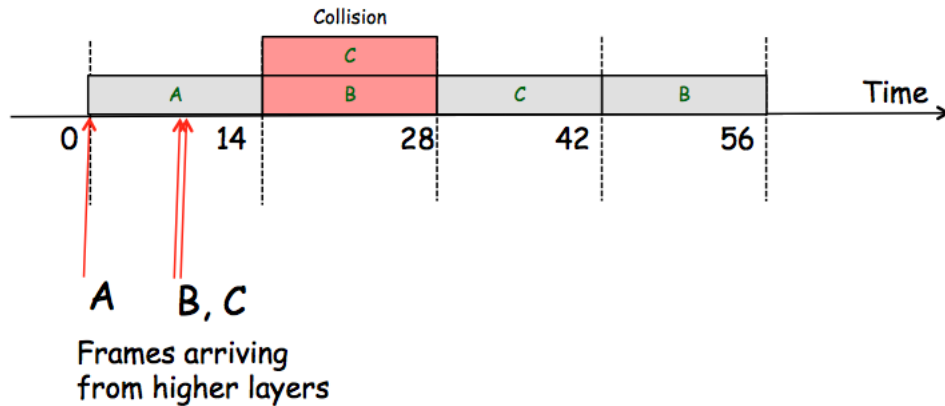
- Station A draws numbers: 0.45, 0.11, 0.71....
- Station B draws numbers: 0.10, 0.65, 0.91....
- Station C draws numbers: 0.83, 0.41, 0.25

The numbers must be used in the sequence they are provided and each random number must be used only once. You can decide to use these random numbers to make randomized decisions as you like, as long as you state your rule. A commonly used rule is the following: given a random number $r \in [0, 1]$, if you want to choose among N points, the point chosen is the k th point ($1 \leq k \leq N$) where $k = \lceil rN \rceil$. (Example 1: if you want to flip a fair coin you can use the random numbers as follows: $r \leq 0.5$ means Heads, $r > 0.5$ means tails. Example 2: if you try to pick one slot randomly out of two slots total: same, as before. Example 3: if you try to pick one slot randomly out of four slots total, you can do the following: $0 \leq r < 0.25$, $0.25 \leq r < 0.5$, $0.5 \leq r < 0.75$, $0.75 \leq r < 1$ can correspond to the first, second, third or fourth timeslot.) You can use the same random numbers for both questions (a) and (b).

- (a) (9 Points) Consider that nodes run a **Slotted Aloha** protocol, where the time slot equals the fixed duration of each frame = 14 time units; i.e., slots start at time 0, 14, 28, ... etc. Show all transmissions that take place according to the Slotted Aloha Protocol, until all three frames are successful.

Answer: B and C start transmitting in the next slide they arrive. This will result in a collision in the second slot. C flips a coin for transmitting in the next slide and succeeds. B on the other hand fails in the first try and succeeds in the second try. Therefore it will transmit in 2 slots after collision.

- (b) (9 Points) Consider a **CSMA/CD** protocol with negligible propagation delay. This means that all stations can detect very quickly whether there is a collision, let's say



within one time-unit, which defines the contention slot (“mini-slot”) for CSMA/CD to be one time unit. Frames are still 14 time units long.

As specified in CSMA (repeated here for your convenience), in a given mini-slot:

- If no host has seized the channel, then all hosts can compete. If exactly one host transmits, then this host will continue until it completes its frame (1+13 minislots long). If two or more nodes transmit, all colliding nodes detect the collision by the end of the minislot and re-schedule their transmission to another minislot using *exponential backoff*. If no host transmits, the channel remains idle in that mini-slot.
- If some host has already seized the channel, all other hosts sense that and refrain from transmitting until the end of transmission.

Answer:

At the beginning of the second slot both B and C sense the idle channel and start transmitting. Quickly, in the first mini slot they will sense the collision. They would try to retransmit in the following mini slots with exponential backoff(0 or 1). B will choose the next minislot and will start retransmission. C will choose 1 in exponential back off and choose the second mini slot. But before transmitting it will sense that B is using channel so it will wait for B to finish his transmission then starts transmitting at time 29.

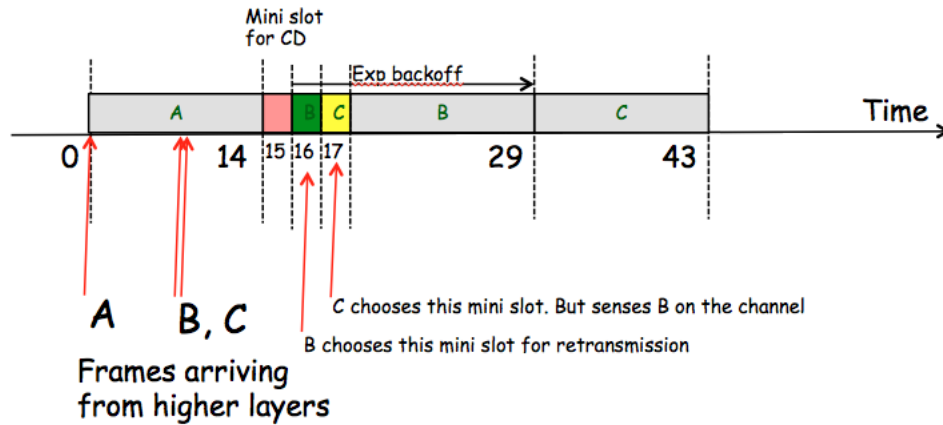
Question: show all transmissions that take place according to CSMA/CD until all frames are successful.

- (c) (2 Points) For this particular scenario, which protocol had the higher throughput: Slotted Aloha or CSMA/CD?

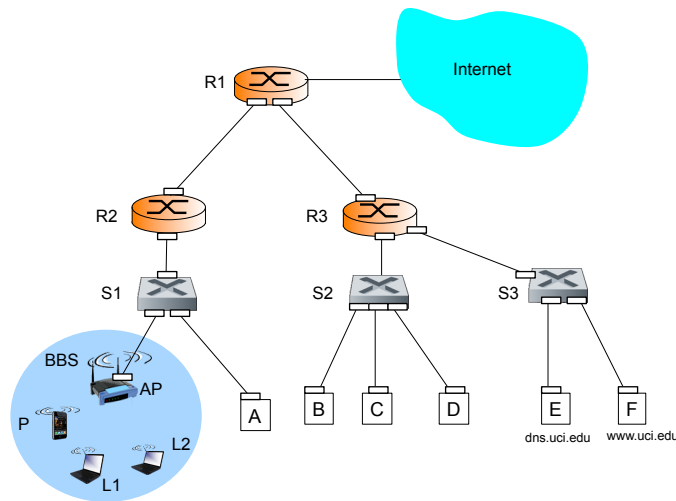
Answer:

Slotted Aloha: $\frac{3 \cdot 14}{56}$

CSMA/CD: $\frac{3 \cdot 14}{43}$



3. (30 Points) **Routers and Switches.** Consider the following figure, depicting 6 desktops (A,B,C,D,E,F), 3 switches (S1, S2, S3), 3 routers (R1, R2, R3) and one BSS. The BSS consists of a WiFi Access Point (AP), 2 laptops (L1, L2) and one smartphone (P).



- (a) (10 Points) IP addresses:
- How many subnets are there? Assign IP addresses to all interfaces using subnets of the form 192.168.1.xxx, 192.168.2.xxx, etc.
 - Show the routing tables at each router, R1, R2, R3.
 - When laptop L1 joins the BSS, describe the DHCP messages exchanged.
- Note: For question (a)(iii), you do not need to describe layer-2 headers.
- (b) (10 Points) Assume that all ARP tables are empty. Describe all MAC frames sent on the network (for each frame indicate the source MAC, dest MAC, src IP, dest IP) when:
- B sends a packet to D.
 - D sends a packet to B.

- (c) (10 Points) Assume that L1 knows the IP address of the web server (F), has already set up a TCP connection and wants to send an HTTP request. Describe all the MAC frames sent on the network (including at least the source MAC, dest MAC, src IP, dest IP, and also some higher layer information) in the following cases:
- L1 sends an HTTP request to F.
 - F sends an HTTP response back to L1.

Note: For questions (b) and (c), when you describe the frames, it is ok to refer to the IP and MAC addresses of an interface with a short name, e.g. IP_A , MAC_A for A, etc.).

Answer:

(A)

- 5 subnets (without considering the WAN connection to Internet).

Subnet 1: (R1-R2), 192.168.4/24

Subnet 2: (R2-S1-A-BSS), 192.168.5/24

Subnet 3: (R1-R3), 192.168.1/24

Subnet 4: (R3-S2-B-C-D), 192.168.2/24

Subnet 5: (R3-S3-E-F), 192.168.3/24

IP range assignment is arbitrary and you may use any range in 192.168.1.xxx, 192.168.2.xxx, etc. Each subnet has 256 hosts and you may assign an IP address to each interface in a subnet.

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R1

192.168.0/22 to subnet3 interface

192.168.4/23 to subnet1 interface

192.168.0/24 to Internet interface

Others to Internet interface

R2

192.168.4/24 to subnet2 interface

Others to subnet1 interface

R3

192.168.2/24 to subnet4 interface

192.168.3/24 to subnet5 interface

Others to subnet1 interface

- Connecting laptop (L1) needs its IP address, addr of first-hop router, addr of DNS server:

DHCP request encapsulated in UDP, encapsulated in IP, encapsulated in a link frame

The frame broadcast on LAN, received at DHCP server

The frame demuxed to IP demuxed to DHCP DHCP server formulated DHCP ACK containing client's IP address, IP address of first hop router for client, name and IP address of DNS

server Encapsulation of DHCP server, frame forwarded to client, demuxing up to DHCP at client

Client now knows its IP address, name and IP address of DSN server, IP address of its first-hop router

(B) In this example, both the source and destination are in the same subnet.

i. B wants to send datagram to D, but D's MAC address not in B's ARP table. B broadcasts ARP query packet, containing:

src MAC: *MAC B*

dest MAC: *FF-FF-FF-FF-FF-FF*

src IP: *IPB*

dest IP: *IPD*

All nodes on LAN receive ARP query as well as D. D replies to B with its *MACD* address, and frame unicast to B, containing:

src MAC: *MACD*

dest MAC: *MACB*

src IP: *IPD*

dest IP: *IPB*

Now the B is able to send the packet to D:

src MAC: *MACB*

dest MAC: *MACD*

src IP: *IPB*

dest IP: *IPD*

ii. D wants to send datagram to B, and B's MAC address is in Ds ARP table from previous step (assuming that ARP query from B updated the ARP table in D). D sends the packet, containing:

src MAC: *MACD*

dest MAC: *MACB*

src IP: *IPD*

dest IP: *IPB*

(C) i.

L1 to AP: (802.11 frame)

Addr1 MAC: *MACAP*

Addr2 MAC: *MACL1*

Addr3 MAC: *MACR2*

src IP: *IPL1*

dest IP: *IPF*

AP to R2: (802.3 frame)

src MAC: *MACL1*

dest MAC: $MACR2$
src IP: IP_{L1}
dest IP: IP^F
R2 to R1:
src MAC: $MACR2$
dest MAC: $MACR1$
src IP: IP_{L1}
dest IP: IP^F

R1 to R3:
src MAC: $MACR1$
dest MAC: $MACR3$
src IP: IP_{L1}
dest IP: IP^F

R3 to F:
src MAC: $MACR3$
dest MAC: MAC^F
src IP: IP_{L1}
dest IP: IP^F

ii.

F to R3:
src MAC: MAC^F
dest MAC: $MACR3$
src IP: IP^F
dest IP: IP_{L1}

R3 to R1:
src MAC: $MACR3$
dest MAC: $MACR1$
src IP: IP^F
dest IP: IP_{L1}

R1 to R2:
src MAC: $MACR1$
dest MAC: $MACR2$

src IP: IP^F
dest IP: IP^{L1}

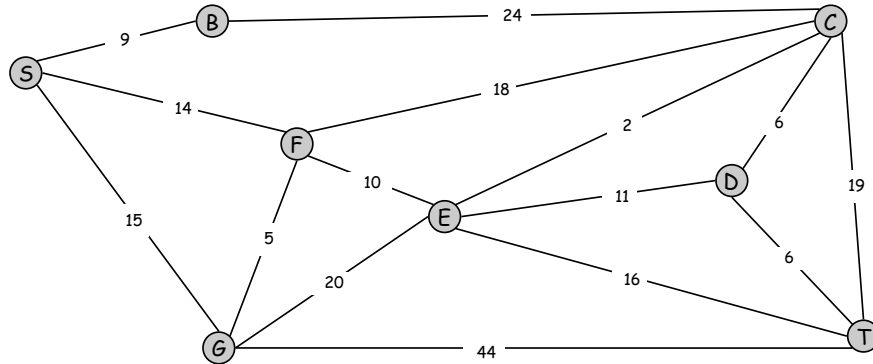
R2 to AP: (802.3 frame)
src MAC: MAC^{R2}
dest MAC: MAC^{L1}
src IP: IP^F
dest IP: IP^{L1}

AP to L1: (802.11 frame)
Addr1 MAC: MAC^{L1}
Addr2 MAC: MAC^{AP}
Addr3 MAC: MAC^{R2}
src IP: IP^F
dest IP: IP^{L1}

SPACE for answer

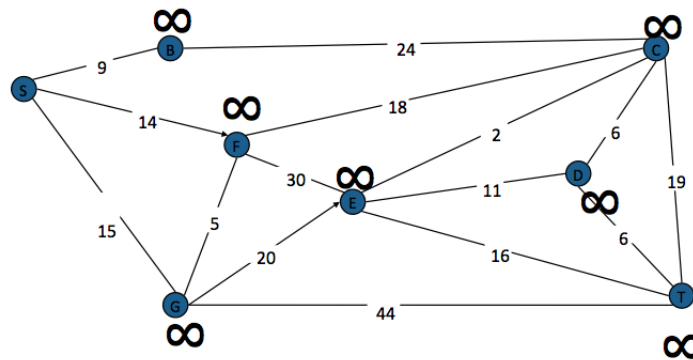
4. (30 Points) Routing Algorithms.

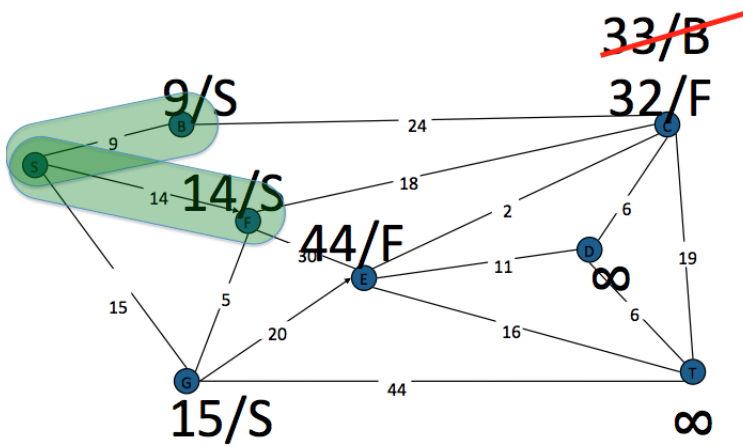
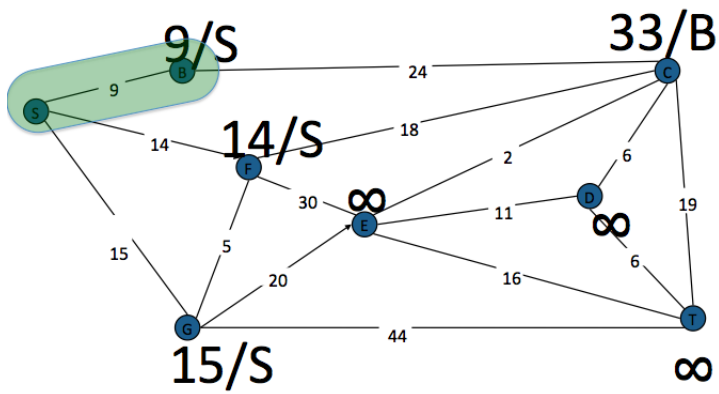
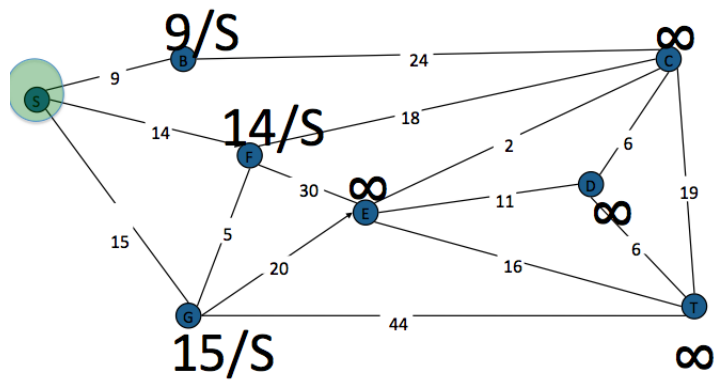
Consider the undirected graph shown on the figure below. The nodes correspond to routers and the edges correspond to links between the routers. There is a cost associated with every edge as indicated on the figure.

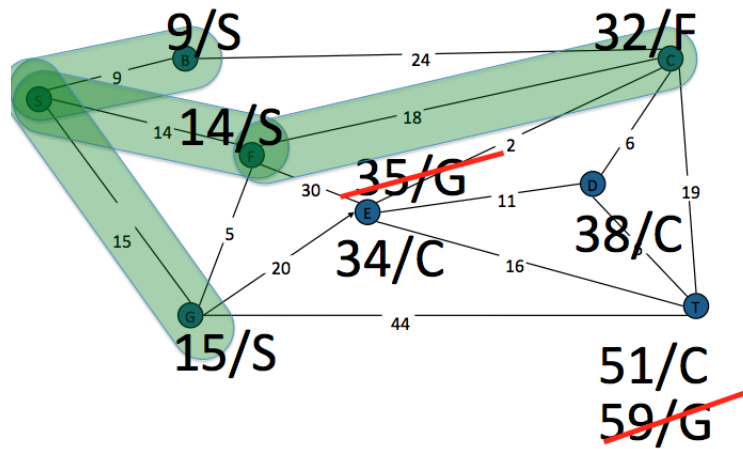
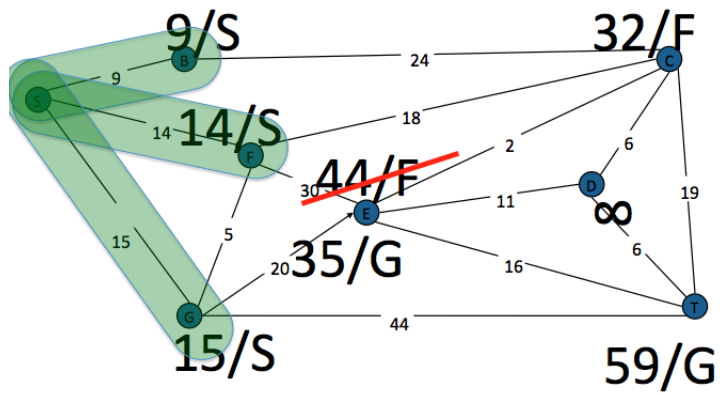


- (15 Points) Find the shortest paths from every node to destination node C . Use one algorithm that you are comfortable with (Dijkstra, Bellman-Ford asynchronous or synchronous). State which algorithm you chose and show your calculation on page 8. At the end, show: (i) the shortest path tree from every node to C and (ii) the entry that every node has in the routing table for destination C : $(C, \text{next hop}, \text{distance})$ (i.e., the shortest path to destination C is “distance” away and via node “next hop”).
- (3 Points) If you use a synchronous version of distance-vector, what is the maximum number of iterations before the distributed algorithm converges? (Notes: You can answer that question without running the algorithm. In each “iteration”, nodes exchange their distance vectors. The algorithm starts with all nodes having only information about their costs to their nearest neighbors.)
- (12 Points) After the algorithm has converged, the link between C and E changes cost from 2 to 1000. Use Bellman-Ford to compute the changes in the shortest paths, if any. Show your calculation and the end result (as in question (a) above) on page 9.

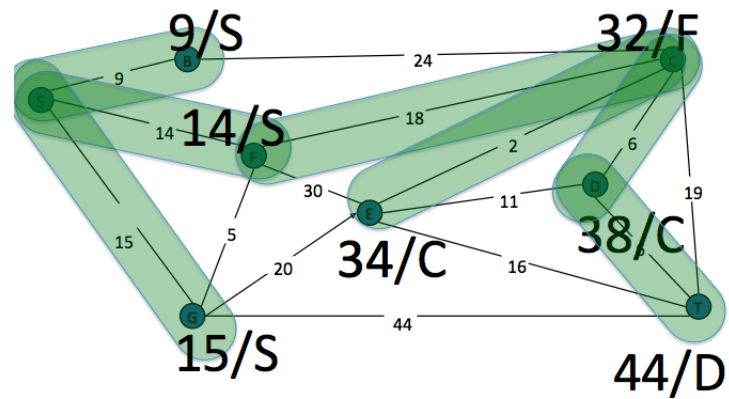
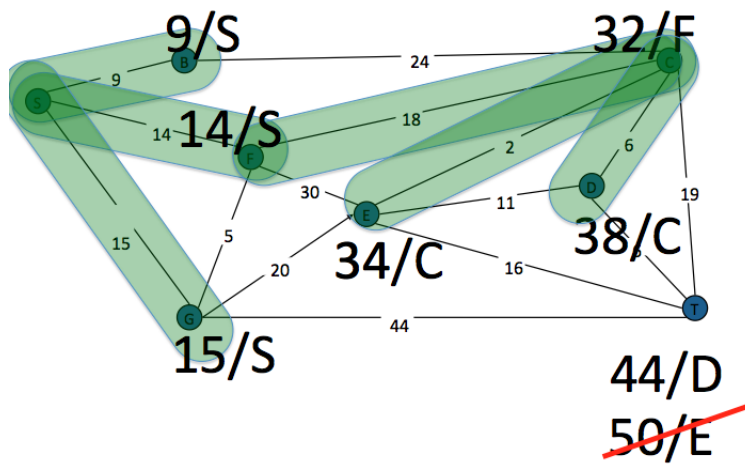
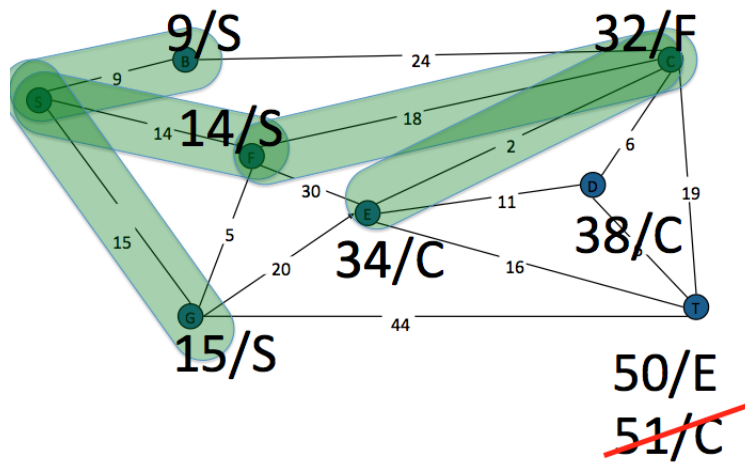
Answer: Here is a summary of solution for both Dijkstra and Bellman-Ford. For complete animations refer to week8 of discussion slides. First set of figures show Dijkstra algorithm implemented on this graph. The following figures show Bellman-Ford algorithm.







ADDITIONAL SPACE



	1	2	3	4	5	6
S	0	0	0	0	0	0
B	Inf.	9/S	9/S	9/S	9/S	9/S
F	Inf.	14/S	14/S	14/S	14/S	14/S
G	Inf.	15/S	15/S	15/S	15/S	15/S
C	Inf.	Inf.	32/F	32/F	32/F	32/F
E	Inf.	Inf.	35/G	34/C	34/C	34/C
D	Inf.	Inf.	Inf.	38/C	38/C	38/C
T	Inf.	Inf.	59/G	51/(C or E)	44/D	44/D

Figure 1: iterations of bellman-Ford

•Shortest paths :

- S – B : 9 (S-B)
- S – F : 14 (S-F)
- S – G : 15 (S-G)
- S – C : 32 (S-F-C)
- S – E : 34 (S-F-C - E)
- S – D : 38 (S-F-C – D)
- S – T : 44 (S-F-C – D - T)

Figure 2: shortest path

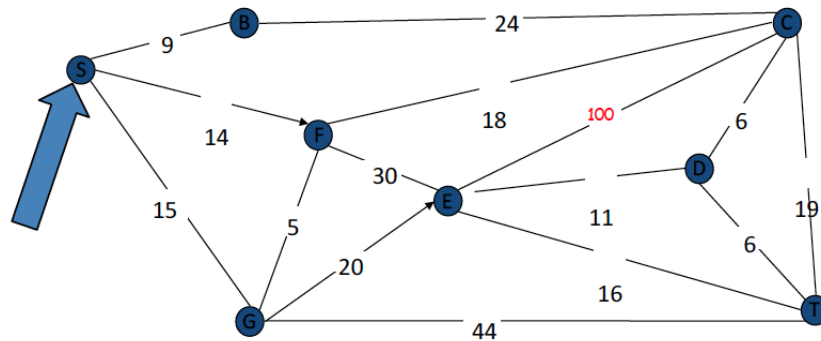


Figure 3: link change value

	Previous	change	1	2
S	0	0	0	0
B	9/S	9/S	9/S	9/S
F	14/S	14/S	14/S	14/S
G	15/S	15/S	15/S	15/S
C	32/F	32/F	32/F	32/F
E	34/C	132/C	35/G	35/G
D	38/C	38/C	38/C	38/C
T	44/D	44/D	44/D	44/D

Figure 4: update of Bellman-Ford based on link value change